INCH-POUND
MIL-PRF-32052/1(CR)
AMENDMENT 1
6 October 2000

PERFORMANCE SPECIFICATION SHEET

BATTERY, RECHARGEABLE, SEALED, BB-XX90/U, BB-X590/U and BB-390/U

This amendment forms a part of MIL-PRF-32052/1(CR), dated 7 December 1999, and is approved for the Communications-Electronics Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title, delete and substitute:

"BATTERY, RECHARGEABLE, SEALED, BB-XX90/U, BB-X590, and BB-390B/U"

PAGE 3

Rated Capacity, delete in its entirety and substitute:

"Rated Capacity at 2A:
BB-XX90: 4.6 AH
BB-X590: 6.0 AH
BB-390B/U: 4.8 AH"

PAGE 4

Delete in its entirety and replace with the attached page.

Custodian:
Army – CR

Preparing Activity:
Army - CR

(Project 6140-A923-001)

AMSC N/A

FSC 6140

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| Discharge Rates | 5 Second Voltage(V) | Final Voltage(V) | Current (Amps) |
|---|---|---|---|
| C/5 | - | 20.0 | 0.8 |
| $0.5C_5$ | - | 20.0 | 2.0 |
| $2.5\ C_5$ | 22.0 | 20.0 | 10.0 |
| $4.5C_5$ (12V) | 11.0 | 10.0 | 36.0 |

| Test | Current or Power (Amperes or Watts) | BB-XX90 Minimum Capacity Requirement (AH or WH) | BB-X590 Minimum Capacity Requirement (AH or WH) | BB-390B/U Minimum Capacity Requirement (AH or WH) |
|---|---|---|---|---|
| Full Capacity Discharge | 2A<br>50W | 4.6AH<br>132 WH | 6.0AH<br>178 WH | 4.8 AH<br>118 WH |
| Cycle Life, 224 cycles | 2A | 4.2 AH | 5.4 AH | 4.3 AH |
| Overcharge, 24 hours | 2A | 4.6 AH | 6.0 AH | 4.8 AH |
| High Rate Discharge | 10A | 4.3 AH | 5.2 AH | 3.0 AH |
| Low Temperature Discharge at -20°C for BB-390B, BB-XX90 and -30°C for BB-X590 (see note 4) | 2A | 4.0 AH | 4.0 AH | 3.3 AH |
| Retention of Charge 7 days at 50° C (122°F) | 2A | 4.0 AH | 5.4 AH | 1.8 AH |
| Pulse Discharge 36 amps, 5 seconds on, 25 seconds off, continuously to 10 volts | 36A | 9.0 AH | 11.4 AH | 8.0 AH |
| Vibration | 0.8A | 4.8 AH | 6.0 AH | 4.9 AH |

NOTES -
1. Unless otherwise specified, all values are minimum.
2. All charges and discharges shall be performed on fully assembled batteries through the terminals.
3. All discharge currents are fixed at the specific rate regardless the increase of battery capacity.
4. Allow discharge voltage to 18.0V.



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